THERMOROUTE HIGHLIGHTS

PRE-OPEN / INCOMPLETE - DESIGN HIGHLIGHTS ONLY

Evidence state: Canonical computation and target-period evaluation incomplete

Use: Internal review; not a performance summary

> Draft status: the canonical computation and one-time target-period evaluation remain incomplete. These are design highlights, not performance findings.

- Is designed to hindcast daily river water temperature at 1-, 3-, and 7-day horizons.
- Anchors a learned temporal model to damped persistence with a bounded residual.
- Uses a stable 120-site USGS development panel covering 2006–2020.
- Separates training, validation, calibration, and exploratory development years.
- The frozen protocol specifies six primary model types and seven one-factor controls.
- Uses station-balanced RMSE effects on identical model-pair target keys.
- Enumerates whole-HUC2 signs and adjusts exactly five formal p-values with Holm.
- The opening design will archive raw target requests, identifiers, qualifiers, and conflicts.
- Requires a model-freeze commit before candidate metadata and later covariates.
- Generates formal statements only from a fully verified one-time receipt.

Current status

The legacy headline numbers have been withdrawn because they belong to a different station mapping and do not have the lineage sidecars required by the current code. The outstanding work is to rerun the canonical development chain, freeze and replay all model bundles, acquire outcome-free metadata and historical covariates in the required order, pass authorization, execute the fixed opening, and regenerate the manuscript and release evidence.

Scope

The architecture is physics-inspired but remains a statistical predictor. Its event threshold and numerical comparison margin are methodological quantities. The external-site analysis uses local observation history and remains exploratory. The current Git seal is repository-internal and has no independent timestamp or custodian.